

THIS IS A GRADED ASSIGNMENT! Answers must be entered in MOODLE before the due date/time.

PROBLEM #1.

You have sampled a population in which you know that the percentage of the homozygous recessive genotype (aa) is 36%. Using that 36%, calculate the questions below. **NOTE: All of your answers should be expressed as a percentage using one decimal place. For example, a HW value of 0.183 would be 18.3% as an answer. The value 0.183 is a relative frequency, whereas 18.3% is a percentage. While they seem the same, MOODLE will only accept a certain format for answers as it deems correct for an answer. ALL answers MUST have one decimal place. For example, an answer of 10% would need be entered as "10.0". Do NOT include the % symbol. Warning: Many people will not get credit for answers simply because they do not read and follow the instructions.**

- The percentage of the "aa" genotype.
- The percentage of the "a" allele.
- The percentage of the "A" allele.
- The percentages of the genotypes "AA" and "Aa."
- The percentages of the two possible phenotypes if "A" is completely dominant over "a."

PROBLEM #2.

Cystic fibrosis is a recessive condition that affects about 1 in 2,500 babies in the Caucasian population of the United States. Please calculate the questions below. **NOTE: All of your answers should be expressed as a percentage using one decimal place. For example, a HW value of 0.183 would be 18.3% as an answer. The value 0.183 is a relative frequency, whereas 18.3% is a percentage. While they seem the same, MOODLE will only accept a certain format for answers as it deems correct for an answer. ALL answers MUST have one decimal place. For example, an answer of 10% would need be entered as "10.0". Do NOT include the % symbol. Warning: Many people will not get credit for answers simply because they do not read and follow the instructions.**

- The percentage of the recessive allele in the population.
- The percentage of the dominant allele in the population.
- The percentage of heterozygous individuals (carriers) in the population.

PROBLEM #3.

In a given population, only the "A" and "B" alleles are present in the ABO system; there are no individuals with type "O" blood or with O alleles in this particular population. If 200 people have type A blood, 50 have type AB blood, and 50 have type B blood, what are the allelic frequencies (expressed as a percentage) of each allele in this population (i.e., what are p and q)? **NOTE: You can still use the HW formula here, but in this case "p" and "q" are CO-DOMINANT alleles, meaning they carry equal dominance and work together to form the blood type "AB". All of your answers should be expressed as a percentage using one decimal place. For example, a HW value of 0.183 would be 18.3% as an answer. The value 0.183 is a relative frequency, whereas 18.3% is a percentage. While they seem the same, MOODLE will only accept a certain format for answers as it deems correct for an answer. ALL answers MUST have one decimal place. For example, an answer of 10% would need be entered as "10.0". Do NOT include the % symbol. Warning: Many people will not get credit for answers simply because they do not read and follow the instructions.**

Your Answers must be uploaded to the link in MOODLE for credit, but you can keep track of them in the table below.

1A: The percentage of the "aa" genotype	
1B: The percentage of the "a" allele	
1C: The percentage of the "A" allele	
1D: The percentage of the genotype "AA" (or homozygous dominant)	
1D: The percentage of the genotype "Aa" (or heterozygous)	
1E: The percentage of the dominant phenotype (or "A_")	
1E: The percentage of the recessive phenotype (or "aa")	
2A: The percentage of the recessive allele in the population	
2B: The percentage of the dominant allele in the population	
2C: The percentage of heterozygous individuals (carriers) in the population	
3A: The percentage of the "A" allele in the population (or "p" for the HW formula)	
3B: The percentage of the "B" allele in the population (or "q" for the HW formula)	